



Participant Information Sheet

Stakeholder Awareness Survey on the Z-Score System

According to the University Grants Commission (UGC), the Z-score method was officially implemented for state university admissions starting from the year 2001. The UGC introduced this system to address issues with the previous raw mark system.

The UGC stated that examination papers naturally vary in difficulty across different subjects, meaning students consistently score higher marks in some fields than in others. For example, a student applying for Biological Sciences could offer either Physics or Agriculture alongside Chemistry and Biology. It was observed that more students obtained higher marks in Agriculture than in Physics. The UGC noted that simply choosing a higher-scoring subject gave students an unfair competitive advantage, which required a standardized correction.

To address this imbalance, the UGC introduced the Z-score method. The Z-score for a student's i -th subject is calculated using the following statistical formula:

$$Z_i = \frac{X_i - \mu_i}{\sigma_i}$$

where

- Z_i = Z-score for the i -th subject (e.g., Z_1 , Z_2 , and Z_3)
- X_i = Raw mark obtained by the student for the i -th subject
- μ_i = Mean (average) mark of all students for the i -th subject

- σ_i = Standard deviation of marks for the i -th subject

The final Z-score of a student is calculated by averaging the Z-scores obtained for the three subjects:

$$\text{Final Student's Z-score} = \frac{Z_1 + Z_2 + Z_3}{3}$$

The primary objective of this survey is to collect and analyze the level of awareness regarding the Z-score method among key educational stakeholders. This includes students, undergraduates, teachers, lecturers, and parents whose children have either recently completed or will soon sit for their A/L examinations.

The statements presented in Part B are not arbitrary viewpoints. They have been compiled from the core arguments and insights of prominent statisticians, educational researchers, and academic critics regarding the Z-score method, drawn directly from published newspaper articles, academic blogs, and research papers. This study aims to understand the opinions of stakeholders about these statements.

The findings from this data analysis will be published as a reference for anyone interested in education. We strictly respect your privacy: all data will be kept confidential, and no personal details of any respondent will be published or shared. Therefore, please feel free to share your honest feedback and support us in this survey.

For further information, please contact:

Principal Investigator: Y.M. Amali P. Rajapaksha

Email: as2022642@sci.sjp.ac.lk

Supervisor:

B.M.S.G. Banneheka

Email: sarath.banneheka@gmail.com